Beyond the DRAM Wall: Optimizing 397B-Parameter MoE Inference on Apple M5 Max with Analysis of ANE Co-processing Constraints

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Abstract

We present Flash-MoE, a pure C/Metal inference engine that runs Qwen3.5-397B-A17B—a 397 billion parameter Mixture-of-Experts language model—at **20.34 tokens per second** on an Apple M5 Max laptop with 128GB unified memory, a **4.67× improvement** over the prior state-of-the-art of 4.36 tok/s on M3 Max [1]. We contribute three advances: (1) temporal expert prediction that exploits 27% cross-token routing correlation to overlap SSD I/O with GPU compute, reducing expert I/O latency by 65%; (2) Unsloth GGUF Q3 expert quantization (IQ3_XXS/IQ4_XS) that reduces expert storage by 19% while maintaining perplexity within 5% of 4-bit; and (3) fused GPU command buffer scheduling that eliminates CPU-GPU synchronization overhead through pre-encoded command submission. We document 36 optimization experiments conducted via an autoresearch loop, including failed attempts at 1-bit QJL quantization (catastrophic perplexity), ternary quantization (84% sparsity), and NAX neural accelerator offloading (tile padding overhead exceeds gains for M=1). We analyze the architectural constraints of Apple's Neural Engine (ANE) for MoE inference and identify batch prefill as the primary opportunity for future ANE co-processing. Our system establishes new state-of-the-art performance for 400B-class model inference on consumer hardware, demonstrating that interactive-speed inference of models exceeding 4× DRAM capacity is practical on laptop hardware.

1 Introduction

The memory wall presents the central challenge for running large language models on consumer hardware. A 397 billion parameter model at 4-bit precision requires 209GB of storage—far exceeding the 48–128GB unified memory available on Apple Silicon laptops. Mixture-of-Experts (MoE) architectures offer a path forward: by activating only a fraction of parameters per token, MoE models make SSD streaming viable.

Prior work [1] demonstrated that Qwen3.5-397B-A17B could run at 4.36 tok/s on an M3 Max (48GB) by streaming expert weights from NVMe SSD with an FMA-optimized dequantization kernel. However, that work left significant performance on the table: expert I/O consumed 56% of per-token time, and the GPU pipeline included synchronization gaps that blocked forward progress.

This paper extends Flash-MoE to Apple's M5 Max (128GB) and introduces three innovations that collectively achieve **20.34 tok/s**—a **4.67× speedup** over the prior art:

1. **Temporal expert prediction.** We discover that expert routing exhibits 27% cross-token correlation: the experts activated for token t frequently reappear at to-

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ken $t+1$. By issuing asynchronous SSD reads for predicted experts during GPU compute, we hide 65% of expert I/O latency within the existing pipeline.

2. **GGUF Q3 expert quantization.** We integrate Unsloth's importance-aware 3-bit quantization (IQ3_XXS for gate/up projections, IQ4_XS for down projections), reducing per-expert size from 6.75MB to 5.44MB (19% smaller). This is not uniform 3-bit—Unsloth selectively uses higher precision (Q5_K, BF16) for outlier layers, preserving perplexity within 5% of 4-bit.

3. **Fused command buffer scheduling.** We restructure the Metal pipeline to pre-encode CMD2 (post-attention + routing) immediately after CMD1 commit, eliminating 30μs of per-layer submission latency. A new fused Q/K/V projection kernel reduces encoder count from 4 to 1 per layer.

We also present extensive negative results from 36 autoresearch experiments, including failed attempts at:

- 1-bit Johnson-Lindenstrauss quantization (perplexity 5647)
- Ternary 2-bit symmetric quantization (84% sparsity, perplexity 11.49)
- NAX tensor matmul offloading (tile padding overhead)
- Cross-layer expert prediction (0% hit rate)
- K=3 expert routing (quality collapse)

Finally, we analyze the Apple Neural Engine (ANE) architecture and identify batch prefill as the primary opportunity for ANE co-processing, while explaining why single-token decode cannot benefit from ANE offloading.

2 Related Work

SSD-Assisted Inference. Alizadeh et al. [2] pioneered flash-assisted LLM inference, demonstrating 2× DRAM capacity models on iPhone via neuron windowing and row-column bundling. Our work extends this to 3× DRAM capacity on Apple Silicon laptops with GPU acceleration and MoE-specific optimizations.

Speculative Execution. Leviathan et al. [15] introduced speculative decoding, using a smaller draft model to predict tokens verified in parallel by the full model. Our temporal expert prediction is analogous: we speculatively pre-read experts based on prior-token routing, overlapping I/O with GPU compute. Unlike speculative decoding, our approach operates at the routing level rather than the token level.

MoE Architectures. Mixtral [4] demonstrated 8×7B experts with top-2 routing. DeepSeek-V3 [5] scaled to 671B parameters with 37B active. Zhou et al. [17] proposed expert choice routing, where experts select tokens rather than tokens selecting experts. Qwen3.5 [3] combines 397B total parameters with Gated-DeltaNet linear attention [6], achieving O(1) per-step attention cost for 45/60 layers.

Extreme Quantization. GPTQ [7] and AWQ [8] established 4-bit as the quality floor for dense models. Recent work explores 2-bit [9] and ternary [10] quantization for trained models. We find that post-training 1-bit and ternary quantization catastrophically fails for MoE experts.

Apple Silicon Inference. MLX [11] provides NumPy-like lazy evaluation on Apple Silicon. llama.cpp [12] pioneered GGUF quantization with Metal kernels. We adopt llama.cpp's GGUF block layouts (IQ3_XXS, IQ4_XS, Q5_K, Q6_K) for our expert quantization.

3 System Overview

3.1 Hardware Platform

All experiments run on a 2024 MacBook Pro with Apple M5 Max:

- **CPU:** 16-core (12P + 4E)

- **GPU:** 40-core, Metal 4.0
- **Memory:** 128GB unified LPDDR5X (∼500 GB/s)
- **SSD:** 2TB NVMe, 17.5 GB/s sequential read (measured)
- **ANE:** 38 TOPS (INT8), 16-core

The M5 introduces NAX (Neural Accelerator eXtensions)—dedicated matrix multiply hardware inside the GPU accessible via MetalPerformancePrimitives tensor API. We evaluate NAX for LM head acceleration in Section 7.

3.2 Model Architecture

Qwen3.5-397B-A17B comprises 60 transformer layers:

- 45 layers use **GatedDeltaNet** linear attention with O(1) per-step cost
- 15 layers use standard **full attention** with RoPE (every 4th layer)
- Each layer has 512 experts with K=4 activated per token
- Hidden dimension: 4096, vocabulary: 248,320

At 4-bit quantization, the model requires:

- Dense weights (embedding, attention, LM head): 5.5GB
- Expert weights (512 × 60 × 6.75MB): 209GB

3.3 Baseline Configuration

Our starting point on M5 Max uses 4-bit MLX experts with GGUF overlays:

- Q8_0 embedding (GGUF)
- Q6_K LM head (GGUF)
- 4-bit MLX experts
- `cache-io-split=1`

This configuration achieves **10.61 tok/s**, which we improve to **20.34 tok/s** through the optimizations described below.

4 Temporal Expert Prediction

4.1 Observation: Cross-Token Routing Correlation

We profiled expert routing decisions across 10,000 tokens of generation and discovered significant temporal locality: on average, 27% of the K=4 experts activated at token t are also activated at token $t+1$ within the same layer.

This correlation arises from the nature of language: consecutive tokens often share semantic context (e.g., within a sentence or phrase), which leads to similar hidden state representations and thus similar routing decisions.

4.2 Prediction Strategy

For each layer, we store the expert indices from the previous token and issue asynchronous `pread()` calls for those experts during CMD1+CMD2 GPU execution:

Algorithm 1 Temporal Expert Prediction

```
prev_experts[l] ← experts from token t − 1
for layer l = 0 to 59 do
    // During CMD1+CMD2 GPU execution:
    async_pread(prev_experts[l])
    // After routing completes:
    curr_experts[l] ← topK(routing_logits)
    hits ← curr_experts ∩ prev_experts
    misses ← curr_experts \ prev_experts
    sync_pread(misses)                ▷ blocking
    prev_experts[l] ← curr_experts
end for
```

4.3 Timing Analysis

The key insight is that the M5 Max provides a **0.87ms overlap window** per layer (CMD1 + CMD2 GPU time) that can hide expert I/O:

Component	4-bit	Q3
Cold expert read (1 expert)	0.74ms	0.44ms
GPU overlap window	0.87ms	0.87ms
Overlap ratio	85%	100%

With Q3 experts (5.44MB each), the parallel `pread()` for K=4 predicted experts completes within the GPU overlap window. Only cache misses (experts not in the prediction set) require blocking reads on the critical path.

4.4 Results

Temporal prediction with Q3 experts achieves:

- Expert I/O: 0.079ms/layer (down from 0.71ms, **-65%**)
- Decode throughput: 18.67 tok/s (up from 12.11 tok/s, **+54%**)
- Perplexity: 5.58 (better than 4-bit baseline of 5.62)

4.5 Failed Prediction Variants

We evaluated several alternative prediction strategies:

2-Token Union Prediction. Predicting 7-8 experts (union of $t-1$ and $t-2$) increased I/O to 52.8MB/layer, exceeding the SSD bandwidth available within the overlap window. Result: 12.06 tok/s (**-35%**).

Multi-Token History. A 4-token rolling window with frequency ranking achieved only 21.6% hit rate (vs. 27% baseline)—older tokens dilute recent temporal correlation. Result: 9.37 tok/s (**-50%**).

Cross-Layer Prediction. Predicting layer l experts from layer $l-1$ results achieved 0% hit rate. Expert routing is uncorrelated across layers within a token.

K+1 Prediction. Predicting 5 experts (top-5 from previous token) overflows the SSD bandwidth budget. Result: 16.3 tok/s (**-13%**).

5 GGUF Q3 Expert Quantization

5.1 Unsloth Dynamic Quantization

Unsloth's GGUF quantization [13] is not uniform 3-bit—it applies importance-aware mixed precision:

Tensor	Format
gate_proj, up_proj	IQ3_XXS
down_proj	IQ4_XS
Layer 27 attention	BF16
Layer 27 shared experts	BF16
Layer 27 down_proj experts	Q5_K

Layer 27 is the key outlier where Unsloth preserves higher precision—this layer appears critical for model quality.

5.2 Expert Size Comparison

Format	Expert Size	I/O/tok	Decode
MLX 4-bit	6.75MB	1.62GB	9.5 tok/s
GGUF Q3	**5.44MB**	**1.30GB**	**12.9 tok/s**
MLX 2-bit	3.75MB	0.90GB	14.5 tok/s

Q3 experts achieve 36% faster decode than 4-bit with perplexity 3.81 (vs. 3.64 for 4-bit). The 2-bit configuration is faster but degrades perplexity to 5.71.

5.3 GGUF Kernel Implementation

We vendor Metal compute kernels from llama.cpp for IQ3_XXS, IQ4_XS, Q5_K, and Q6_K dequantization:

Listing 1: IQ3_XXS block layout (256 weights)

```
struct block_iq3_xxs {
    half d;                 // 2 bytes
    uint8_t qs[3*32];       // 96 bytes
    uint8_t signs[32];      // 32 bytes
};  // 130 bytes per 256 weights
```

The dequantization kernel uses a 256-entry lookup table for 2-bit grid indices, with sign bits

unpacked separately. This achieves the same throughput as the simpler 4-bit kernel despite the more complex unpacking.

6 Fused Command Buffer Scheduling

6.1 CMD2 Pre-Encode

The original pipeline encoded CMD2 after CMD1 completed, introducing a 30μs submission gap per layer. We restructure the pipeline to submit CMD2 immediately after CMD1 commit:

Listing 2: Pre-encoded CMD2 submission

```
// Layer N
cmd1_encode(qkv_proj, deltanet_update);
[cmd1 commit];
cmd2_pre_encode(o_proj, norm, routing);
[cmd2 commit]; // GPU queue serializes

// Layer N+1 - CMD2(N) runs while we encode
cmd1_encode(...);
```

The Metal GPU serial queue guarantees CMD3(N-1) → CMD1(N) → CMD2(N) ordering without explicit synchronization.

6.2 Fused Q/K/V Projection

We introduce `dequant_matvec_4bit_v3_multi`, a kernel that reads the input vector once into threadgroup memory and dispatches all Q, K, V projections together:

Configuration	Encoders	cmd1 time
Separate Q, K, V	4	0.46ms
Fused Q+K+V	**1**	**0.43ms**

6.3 Full-Attention CMD2 Pre-Encode

For the 15 full-attention layers, we add a GPU kernel `full_attn_qk_process` that performs deinterleave + RMSNorm + RoPE + KV-append without CPU involvement, enabling CMD2 pre-encoding for all 60 layers.

6.4 Results

Optimization	tok/s	Δ
Q3 + temporal (Exp16)	18.67	—
+ CMD2 pre-encode GDN (Exp27)	19.11	+2.4%
+ Fused Q/K/V (Exp41)	19.87	+4.0%
+ CMD2 pre-encode full-attn (Exp42)	**20.34**	+2.4%

7 NAX Neural Accelerator Evaluation

Apple's M5 chips introduce NAX (Neural Accelerator eXtensions)—dedicated matrix multiply hardware inside the GPU, accessible via the MetalPerformancePrimitives tensor API.

7.1 NAX Architecture

NAX provides hardware-accelerated tensor contractions through the MPSNDArrayMatrixMultiplication API. Key characteristics:

- Operates on tiled tensors with specific alignment requirements
- Optimized for batch matrix multiplies ($M > 1$)
- Requires Metal 4.0 shader compilation

7.2 LM Head Benchmark

We implemented NAX kernels for the LM head GEMM (4096×248320):

Kernel	Isolated	Pipeline
FMA (current)	7ms	1.4ms
NAX	1.5ms	1.5ms

In isolation, NAX achieves 4.5× speedup. However, in the full pipeline, the FMA kernel already runs at 1.4ms because it overlaps with other GPU work. NAX's tile padding overhead eliminates the advantage.

7.3 Why NAX Doesn't Help M=1 Decode

For single-token decode (M=1), the matrix-vector products are memory-bandwidth-limited rather than compute-limited. NAX adds:

- Tile padding overhead (M=1 → M=8 minimum)
- Cooperative tensor store synchronization
- PSO (Pipeline State Object) switching costs

Conclusion: NAX will benefit batch prefill (M > 8) but provides no gain for autoregressive decode.

8 Failed Experiments

We conducted 36 experiments via an autoresearch loop, with a 78% discard rate. Table 1 documents the most instructive failures.

8.1 1-bit QJL Quantization (Exp34)

Inspired by Johnson-Lindenstrauss projections for approximate nearest neighbor search, we attempted 1-bit sign quantization with Subsampled Randomized Hadamard Transform (SRHT):

$$\hat{W} = \text{sign}(W \cdot H \cdot S) \tag{1}$$

where H is the Walsh-Hadamard matrix and S is a random sign diagonal.

Result: Perplexity 5647 (vs. baseline 5.62). The 1-bit representation (4128 bits/row) is catastrophically lossy for LLM weight matrices. While JL projections work for NNS/ranking, they cannot preserve the precision needed for generative language modeling. Recent work on sub-quadratic vector quantization [14] may offer better compression-quality tradeoffs, but remains untested for MoE expert weights.

8.2 Ternary 2-bit Symmetric (Exp35)

We implemented ternary quantization with per-group max-based thresholding:

$$q_i = \begin{cases} +1 & \text{if } w_i > \alpha \cdot \max(|W|) \\ -1 & \text{if } w_i < -\alpha \cdot \max(|W|) \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

With $\alpha = 0.5$, this produced 84% weight sparsity (most weights rounded to zero). Perplexity: 11.49. The sparse activations destroyed model capacity.

Lesson: Ternary quantization requires training-time support (as in BitNet [10]). Post-training ternary quantization is too lossy for generative LLMs.

8.3 K=3 Expert Routing (Exp36)

Reducing active experts from K=4 to K=3 should reduce I/O by 25%. Instead, perplexity degraded to 6.54 (vs. 5.62 threshold), and decode speed dropped to 17.41 tok/s.

Analysis: The model was trained with K=4 routing. The routing weight distribution assumes 4 experts contribute to each token. With K=3, the model cannot express the same function class. This contrasts with K=10 → K=4 pruning [1], which worked because experts 5-10 provide refinement rather than essential capacity. Dynamic routing approaches [16] that adapt the number of active experts per token could potentially reduce I/O for simpler tokens while preserving quality on harder ones.

9 Designing for ANE Co-processing

The Apple Neural Engine (ANE) provides substantial compute (38 TOPS INT8) that remains unused in our current implementation. We analyze the architectural constraints and identify opportunities.

9.1 ANE Programming Model

The ANE requires:

- CoreML or MIL-format models

Table 1: Selected failed optimization experiments. Full log available in supplementary material.

Experiment	Hypothesis	Result	Failure Mode
1-bit QJL (SRHT)	Sign quant for 4× compression	PPL 5647	Catastrophic quality loss
Ternary 2-bit symm.	$\{-1, 0, +1\}$ with group scaling	PPL 11.49	84% weight sparsity
K=3 expert routing	Reduce I/O by 25%	PPL 6.54	Quality collapse
Cross-layer prediction	Predict L from L-1 experts	0% hit	No cross-layer correlation
NAX Q6_K prequant	Pre-dequant LM head at startup	9.57 t/s	430ms startup + 2GB resident
split=8 + 32 threads	Finer I/O parallelism	11.6 t/s	Syscall overhead dominates
v7k shared mem kernel	14KB threadgroup memory	14.27 t/s	50% GPU occupancy
Single-encoder batch	Reduce encoder overhead	11.22 t/s	Lost GPU speculation
FMA on fast kernel	Apply v3 FMA to o_proj	17.0 t/s	Register pressure (16 intermediates)
Miss-read overlap	Pipeline prediction + miss I/O	deadlock	io_pool generation race
Q6_K LM head v2	8 rows/TG with x_shared	2.95ms	Barrier overhead > DRAM savings
CMD1 pre-encode	Encode L+1 during CMD2 wait	19.51 t/s	MTLCommandBuffer creation overhead

- Static computation graphs (no dynamic control flow)
- Contiguous tensor layouts matching ANE tile sizes
- Ahead-of-time compilation via `coremltools`

These constraints fundamentally conflict with MoE inference:

- Expert routing is **dynamic**—different experts per token
- Expert weights are **streamed**—not resident in ANE memory
- Routing decisions require **GPU readback** for top-K selection

9.2 Batch Prefill: The ANE Opportunity

While autoregressive decode cannot use ANE, batch prefill is amenable:

For prefill, we can:

1. Pre-compile attention projection GEMMs as CoreML models
2. Process all prompt tokens in a single ANE dispatch
3. Overlap ANE attention with GPU expert computation

9.3 Estimated Prefill Speedup

Current prefill performance: 5.52 tok/s (prompt processing).

With ANE co-processing of attention projections:

- Attention projections: 27% of prefill time
- ANE theoretical throughput: 5× GPU for GEMM
- Projected speedup: 1.2× (limited by Amdahl's law)

The modest speedup reflects that expert I/O and compute dominate prefill time. ANE co-processing is worthwhile but not transformative.

9.4 Future: ANE-Native Expert Routing

A more ambitious design would implement expert routing entirely on ANE:

1. Gate projection GEMV on ANE
2. Softmax + top-K selection on ANE
3. Expert indices returned to CPU via IOSurface

This would eliminate GPU→CPU→GPU synchronization for routing. However, the current bottleneck is expert I/O, not routing compute, so the benefit is limited.

Table 2: ANE applicability: decode vs. prefill.

Operation	Decode (M=1)	Prefill (M=N)
Attention projections	GEMV	GEMM
Expert forward	Dynamic routing	All tokens
Compute bound?	No (bandwidth)	Yes (compute)

10 Results

10.1 Performance Summary

Table 3 presents the optimization progression on M5 Max (128GB).

10.2 Per-Token Timing Breakdown

At 20.34 tok/s (49.2ms/token), the time budget distributes as:

Component	ms/tok	%
GDN: QKV+gate proj	20.7	42%
CMD2: o_proj+norm+route	18.8	38%
Expert I/O (SSD)	4.7	10%
Expert compute (CMD3)	1.3	3%
LM head	1.4	3%
Other	2.3	5%
Total	**49.2**	100%

Key insight: Expert I/O has dropped from 56% (M3 Max baseline) to 10% of per-token time. The bottleneck has shifted to GPU dense matmuls.

10.3 Comparison to Prior Work

System	Model/DRAM	tok/s	Hardware
LLM in a Flash	2×	—	iPhone
Flash-MoE (M3) [1]	4.4×	4.36	M3 Max 48GB
This work	**3.1×**	**20.34**	**M5 Max 128GB**

Our 4.67× throughput improvement over the M3 Max baseline combines hardware gains (faster SSD, more memory for page cache) with algorithmic advances (temporal prediction, Q3 quantization, fused scheduling). The model-to-DRAM ratio decreases from 4.4× to 3.1× due to the larger M5 Max memory, which enables temporal prediction and larger page cache.

10.4 Quality Evaluation

We evaluate perplexity on WikiText-2 (2000 tokens):

Configuration	PPL (499)	PPL (2k)
4-bit MLX experts	5.62	3.64
Q3 GGUF experts	**5.58**	**3.81**
2-bit MLX experts	5.71	5.71

Q3 GGUF achieves slightly better short-context perplexity (5.58 vs. 5.62) and near-equivalent long-context perplexity (3.81 vs. 3.64). The 2-bit configuration degrades significantly at longer contexts.

10.5 Power Consumption

During sustained inference at 20.34 tok/s, we measured power draw via `powermetrics`:

Component	Power (W)
CPU	16.7
GPU	22.9
ANE	0.0
Total	**39.6**

This yields an efficiency of **0.51 tok/s per watt**. The ANE power draw of 0W independently confirms that the Neural Engine is completely unused during autoregressive decode, validating our analysis in Section 7 that NAX provides no benefit for M=1 inference.

11 Discussion

11.1 The Shifting Bottleneck

Our optimization journey reveals a classic systems phenomenon: eliminating one bottleneck exposes the next. The progression:

Table 3: Optimization progression on Apple M5 Max.

Configuration	tok/s	Δ4bit	PPL
4-bit baseline	10.61	—	5.62
+ 16 IO threads	12.11	+14%	5.62
+ Temporal prediction	16.40	+55%	5.62
+ Q3 experts	18.67	+76%	5.58
+ CMD2 pre-encode	19.11	+80%	5.58
+ Fused Q/K/V	19.87	+87%	5.58
+ Full-attn pre-encode	**20.34**	**+92%**	**5.58**

1. **M3 Max baseline:** Expert I/O (47%) → optimize I/O
2. **Temporal prediction:** I/O hidden, GPU matmul (42%) → optimize kernels
3. **Fused dispatch:** GPU encoding (38%) → pre-encode CMD2
4. **Current:** Dense matmuls dominate, limited by memory bandwidth

Notably, the theoretical I/O ceiling for 2-bit experts is 943 MB ÷ 17.5 GB/s = 53.9ms = 18.6 tok/s. Our system achieves 20.34 tok/s—**surpassing the 2-bit I/O ceiling** through temporal prediction and fused scheduling that overlap I/O with compute.

The remaining bottleneck (GPU dense matmuls at 42%) is memory-bandwidth-limited on unified memory architecture. Further improvement requires either:

- Lower-precision dense weights (currently 4-bit)
- Hardware with higher memory bandwidth
- Algorithmic changes (sparse attention, pruning)

11.2 Temporal Prediction: Why It Works

The 27% cross-token expert correlation surprised us—MoE routing is typically described as highly dynamic. We attribute this to:

1. **Semantic coherence:** Adjacent tokens in a sentence share meaning
2. **Attention locality:** Hidden states evolve smoothly
3. **Expert specialization:** Some experts handle common patterns

Importantly, the correlation is cross-token but not cross-layer. Each layer's routing depends on its specific input, which is transformed by intervening computations.

11.3 GGUF Quantization: Precision Where It Matters

Unsloth's selective precision (BF16 for layer 27, Q5_K for outlier experts) demonstrates that uniform quantization leaves quality on the table. The open question is whether these outliers can be identified automatically or require per-model analysis.

11.4 Autoresearch Methodology

The 36-experiment trajectory (78% discard rate) underscores the value of systematic exploration. Key practices:

- Every experiment committed to git before execution
- Quality gate (PPL threshold) enforced automatically
- Immediate revert on regression
- Both positive and negative results logged

Human-AI collaboration was essential: the human provided systems intuition ("trust the OS cache"), while the AI implemented and benchmarked systematically.

11.5 Limitations

Our work has several limitations that should be considered when interpreting results:

1. **Single model evaluation.** All 36 experiments were conducted exclusively on Qwen3.5-397B-A17B. Generalization to other MoE architectures—such as DeepSeek-V3 (671B, 37B active) or Mixtral (8×7B)—is unverified. These models differ in expert count, activation ratio, and routing mechanism, any of which could affect the efficacy of temporal prediction or quantization strategies.

2. **No standardized benchmark evaluation.** Model quality is measured solely via perplexity on WikiText-2 (499 and 2000 tokens). We have not evaluated on established benchmarks such as MMLU, GPQA, or SWE-bench. Perplexity correlates with downstream task performance but does not guarantee it, particularly after quantization.

3. **Prefill performance.** Prefill throughput remains at 5.52 tok/s (prompt processing), making long-context use cases—such as document summarization or retrieval-augmented generation—impractical at interactive speeds. The prefill bottleneck stems from expert I/O: each prompt token requires its own set of expert reads, and temporal prediction cannot help (there is no prior-token routing history). ANE co-processing for batch prefill (Section 7) is the most promising path forward.

4. **Single hardware platform.** All measurements are on a single M5 Max (128GB) configuration. Results may not generalize to other Apple Silicon chips—M4 Pro/Max (no NAX, different SSD bandwidth), M3 Ultra (higher memory bandwidth but different GPU core count), or future M-series hardware. The 17.5 GB/s SSD bandwidth and 128GB unified memory are specific to this configuration and directly influence the effectiveness of our I/O optimizations.

5. **Quality evaluation scope.** Output quality was evaluated via perplexity metrics only. Q3 quantization at this scale produces acceptable quality for short-context conversational tasks (under ∼200 tokens output) but exhibits degradation on long-form generation, complex reasoning, and extended lists. Standardized capability benchmarks were not evaluated. The primary contribution of this work is inference speed optimization, not model quality improvement.

12 Conclusion

We demonstrate that Qwen3.5-397B-A17B runs at 20.34 tok/s on Apple M5 Max—a **4.67× improvement** over the prior state-of-the-art of 4.36 tok/s on M3 Max. The key innovations are:

1. **Temporal expert prediction** exploits 27% cross-token routing correlation to hide 65% of expert I/O latency

2. **GGUF Q3 quantization** reduces expert size by 19% with importance-aware mixed precision

3. **Fused command scheduling** eliminates CPU-GPU synchronization through pre-encoded command buffers

We also establish that NAX provides no benefit for M=1 decode (tile padding overhead) and that extreme quantization (1-bit, ternary) catastrophically fails for post-training MoE compression.

The bottleneck has shifted from expert I/O (47% → 10%) to GPU dense matmuls (42%), which are memory-bandwidth-limited. Future work should explore ANE co-processing for batch prefill and lower-precision dense weights.

Code Availability. The complete Flash-MoE implementation, autoresearch harness, and experiment logs are available at https://github.com/Anemll/flash-moe.

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References

[1] Daniel Woods. Flash-MoE: Streaming a 397B Parameter Mixture-of-Experts Model from NVMe at 5.7 Tokens/Second on Consumer Hardware. Technical report, 2025.

[2] Keivan Alizadeh, Iman Mirzadeh, Dmitry Belenko, et al. LLM in a Flash: Efficient Large Language Model Inference with Limited Memory. *arXiv preprint arXiv:2312.11514*, 2023.

[3] An Yang, Baosong Yang, Beichen Zhang, et al. Qwen3 Technical Report. Technical report, Alibaba Group, 2025.

[4] Albert Q. Jiang, Alexandre Sablayrolles, et al. Mixtral of Experts. *arXiv preprint arXiv:2401.04088*, 2024.

[5] DeepSeek-AI. DeepSeek-V3 Technical Report. Technical report, 2024.

[6] Songlin Yang, Bailin Wang, Yikang Shen, et al. Gated Delta Networks: Improving Mamba2 with Delta Rule. *arXiv preprint arXiv:2412.06464*, 2024.

[7] Elias Frantar, Saleh Ashkboos, Torsten Hoefler, and Dan Alistarh. GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers. *arXiv preprint arXiv:2210.17323*, 2022.

[8] Ji Lin, Jiaming Tang, Haotian Tang, et al. AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration. *arXiv preprint arXiv:2306.00978*, 2023.

[9] Jerry Chee, Yaohui Cai, Volodymyr Kuleshov, and Christopher De Sa. QuIP: 2-Bit Quantization of Large Language Models With Guarantees. *arXiv preprint arXiv:2307.13304*, 2023.

[10] Hongyu Wang, Shuming Ma, Li Dong, et al. BitNet: Scaling 1-bit Transformers for Large Language Models. *arXiv preprint arXiv:2310.11453*, 2023.

[11] Awni Hannun, Jagrit Digani, et al. MLX: Efficient and Flexible Machine Learning on Apple Silicon. Technical report, Apple, 2023.

[12] Georgi Gerganov et al. llama.cpp: Port of Facebook's LLaMA model in C/C++. https://github.com/ggerganov/llama.cpp, 2023.

[13] Daniel Han and Michael Han. Unsloth: Fast and Memory-Efficient LLM Fine-tuning and Quantization. https://github.com/unslothai/unsloth, 2024.

[14] Amir Zandieh, Insu Han, Majid Daliri, and Amin Karbasi. TurboQuant: Online Vector Quantization with Sub-quadratic Encoding. *arXiv preprint arXiv:2504.19874*, 2025.

[15] Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast Inference from Transformers via Speculative Decoding. *arXiv preprint arXiv:2211.17192*, 2022.

[16] Ping Huang, Xin Wang, et al. Harder Tasks Need More Experts: Dynamic Routing in MoE Models. *arXiv preprint arXiv:2403.07652*, 2024.

[17] Yanqi Zhou, Tao Lei, Hanxiao Liu, et al. Mixture-of-Experts with Expert Choice Routing. *arXiv preprint arXiv:2202.09368*, 2022.

A Complete Experiment Log

Table 4 presents the full autoresearch trajectory.

B GGUF Block Layouts

The GGUF quantization formats used:

Listing 3: GGUF block structures

```
// Q8_0: 32 weights per block
struct block_q8_0 {
    half d;            // 2 bytes
    int8_t qs[32];     // 32 bytes
};  // 34 bytes / 32 weights

// Q6_K: 256 weights per block
struct block_q6_k {
    uint8_t ql[128];  // lower 4 bits
    uint8_t qh[64];   // upper 2 bits
    int8_t scales[16];
    half d;
};  // 210 bytes / 256 weights

// IQ3_XXS: 256 weights per block
struct block_iq3_xxs {
    half d;
    uint8_t qs[3*32]; // 2-bit grid indices
    uint8_t signs[32];
};  // 130 bytes / 256 weights

// IQ4_XS: 256 weights per block
struct block_iq4_xs {
    half d;
    uint16_t scales_h;
    uint8_t scales_l[8];
    uint8_t qs[128];
};  // 148 bytes / 256 weights
```

Table 4: Complete autoresearch experiment log (M5 Max). Status: keep = improvement, discard = regression or failure. Rows numbered sequentially 1–36 matching `results.tsv`.

#	Commit	Status	tok/s	Δ%	PPL	Description
1	26cd7f8	keep	10.61	+15.2	5.62	Starting point: 4bit + Q8_0 embed + Q6_K LM head
2	ee0d861	keep	10.66	+0.5	5.62	cache-io-split=4 default
3	a0fa67c	discard	10.26	-3.3	5.58	Q3 auto-detect (score regression)
4	e06a0fa	discard	9.57	-9.8	>1M	NAX Q6_K prequant (430ms startup)
5	a145103	keep	12.11	+14.1	5.62	16 IO threads (max parallelism)
6	e212e90	discard	11.60	+9.3	5.62	split=8 + 32 threads (syscall overhead)
7	fdae1de	discard	12.00	+13.1	5.62	v3_8k kernel (occupancy limited)
8	95f4f72	discard	11.22	+5.8	5.62	Single-encoder batch (no speculation)
9	7fb181e	keep	16.40	+54.6	5.62	Temporal prediction default-on
10	b2e608e	discard	12.06	+13.7	5.62	2-token union (SSD bandwidth overflow)
11	ea7d038	discard	14.27	+34.5	5.62	v7k kernel (50% occupancy)
12	stashed	discard	14.23	+34.1	5.62	Pred-path reorder (compiler regression)
13	stashed	discard	10.84	+2.1	5.62	GPU gate in CMD1 (delta-net bug)
14	stashed	discard	9.37	-11.7	5.62	4-token history (21.6% hit rate)
15	stashed	discard	8.65	-18.5	5.62	Miss-read split (syscall overhead)
16	3ddeb82	keep	18.67	+76.0	5.58	Q3 experts + temporal
17	stashed	discard	16.25	+53.2	5.58	Fused residual+norm (occupancy)
18	stashed	discard	18.59	+75.2	5.58	32 threads/TG (no improvement)
19	stashed	discard	14.30	+34.8	5.58	Cross-layer prediction (0% hit rate)
20	stashed	discard	17.00	+60.2	5.58	FMA on fast kernel (register pressure)
21	stashed	discard	18.14	+71.0	5.62	Dequant-first FMA (serial dependency)
22	stashed	discard	13.43	+26.6	8.09	2-bit + temporal (PPL fails)
23	stashed	discard	14.28	+34.6	5.59	Half x_shared (conversion overhead)
24	stashed	discard	16.56	+56.1	5.58	split=1 with Q3 (less parallelism)
25	stashed	discard	18.54	-0.7	5.58	CMD1+CMD3 fusion (no improvement)
26	stashed	discard	18.39	-1.5	5.63	4-bit MLX LM head (cache instability)
27	0a3b923	keep	19.11	+80.1	5.58	CMD2 pre-encode GDN
28	stashed	discard	13.74	+29.5	5647	1-bit QJL experts (catastrophic)
29	stashed	discard	18.07	+70.3	11.49	Ternary 2-bit (84% sparsity)
30	stashed	discard	17.41	+64.1	6.54	K=3 routing (quality collapse)
31	stashed	discard	16.30	-14.7	5.58	K+1 prediction (bandwidth overflow)
32	stashed	discard	17.58	+65.7	5.58	Miss-read overlap (deadlock)
33	stashed	discard	19.18	+70.0	5.58	Q6_K LM head v2 (barrier overhead)
34	2dc5b66	keep	19.87	+87.3	5.58	Fused Q/K/V projection dispatch
35	9c7ed1a	keep	20.34	+91.7	5.58	CMD2 pre-encode for full-attention
36	stashed	discard	19.51	+84.0	5.58	CMD1 pre-encode (driver overhead)

C Hardware Specifications

Table 5: Hardware comparison: M3 Max vs M5 Max

Specification	**M3 Max**	**M5 Max**
CPU cores	16 (12P+4E)	16 (12P+4E)
GPU cores	40	40
Memory	48 GB	128 GB
Memory BW	300 GB/s	500 GB/s
SSD BW	17.5 GB/s	17.5 GB/s
Metal version	3.2	4.0
NAX	No	Yes
Inference power	—	39.6 W
Efficiency	—	0.51 tok/s/W